a) $\lim_{x \to 3} \frac{\sqrt{2x^2-2}-4}{x+3} = \frac{\sqrt{16}-4}{0} = \frac{0}{0}$ Indet

$$\frac{(\sqrt{2x^2-2}-4)(\sqrt{2x^2-2}+4)}{(x+3)(\sqrt{2x^2-2}+4)}$$

$$\frac{(\sqrt{2x^2-2})^2-16}{(x+3)(\sqrt{2x^2-2}+4)}$$

$$\frac{2x^2-18}{(x+3)(\sqrt{2x^2-2}+4)}$$

$$\frac{2(x^2-9)}{(x+3)(\sqrt{2x^2-2}+4)}$$

$$\frac{2(x-3)\cancel{(x+3)}}{\cancel{(x+3)}(\sqrt{2x^2-2}+4)} = \frac{2(x-3)}{\sqrt{2x^2-2}+4} = \frac{-12}{8} = \boxed{-\frac{3}{2}}$$

b)

b) $\lim_{x \to 2} \dfrac{2\,\text{sen}(x^2-4)}{4x-8} = \dfrac{2 \cdot \text{sen}\,0}{0} = \dfrac{0}{0}$

$$\frac{2\,\text{sen}\left[(x+2)(x-2)\right]}{4(x-2)}$$

$$\frac{2}{4} \cdot \frac{\text{sen}\left[(x+2)(x-2)\right]}{x-2} \cdot \frac{(x+2)}{(x+2)}$$

$$\lim_{x \to 2} (x+2) \cdot \frac{2}{4} \cdot \frac{\text{sen}\left[(x+2)(x-2)\right]}{(x-2)(x+2)} =$$

$$4 \cdot \frac{2}{4} \cdot 1 = \frac{8}{4} = \boxed{2}$$

c) $\lim_{x \to +\infty} \left(1 + \dfrac{\overbrace{3x+1}^{\text{menor}^{\circ}}}{\underbrace{2x^2}_{\text{mayor}^{\circ}}}\right)^{x+2} = (1+0)^{\infty} = 1^{\infty}$

$$\left(1 + \frac{3x+1}{2x^2}\right)^{x+2} = \left(1 + \frac{1}{\frac{2x^2}{3x+1}}\right)^{x+2} =$$

$$\left(1 + \frac{1}{\frac{2x^2}{3x+1}}\right)^{(x+2)\left(\frac{2x^2}{3x+1}\right)\left(\frac{3x+1}{2x^2}\right)}$$

$$\left(1 + \frac{1}{\frac{2x^2}{3x+1}}\right)^{\left(\frac{2x^2}{3x+1}\right) \cdot (x+2)\left(\frac{3x+1}{2x^2}\right)}$$

$$\boxed{e^{3/2}}$$

$$\frac{(3x+1)(x+2)}{2x^2}$$

$$\frac{3x^2+6x+x+2}{2x^2}$$

$$\lim_{x \to +\infty} \frac{\underbrace{3}x^{\overbrace{2}^{\text{grado } 2}}+7x+2}{\underbrace{2}x^{\underbrace{2}_{\text{grado } 2}}} = \frac{3}{2}$$

2) a) $\lim_{x \to -\infty} 2^x = 2^{-\infty} = \frac{1}{2^{\infty}} = \frac{1}{\infty} = 0$

Falso, es 0.

b) $\lim_{x \to +\infty} \left(\frac{\overbrace{2x^1+5}^{\text{grado } 1}}{\underbrace{5x-1}_{\text{grado } 1}} \right)^x = \frac{2}{5}^{\infty} = 0$

Falso. El limite es 0. Al resolver terminamos con $\frac{2}{5}^{\infty}$, entonces como $0 < \frac{2}{5} < 1$ al elevarlo a $+\infty$ el resultado es 0.

c) Falso, factorizar es solo una de las posibles formas, depende de la funcion. Puede usarse el conjugado, operaciones algebraicas, etc.